(9) a) m atoms at a site after time t?

Poisson Distribution

dt is average after time t so

$$P_{dt} = \frac{e^{-dt}(dt)^m}{m!}$$ is the probability for m atoms

fraction not covered?

This corresponds to probability $m=0$

$$P(m=0) = \frac{e^{-dt}(dt)^0}{0!} = e^{-dt}$$

b) variance?

$$\text{variance} = \langle x^2 \rangle_c = \langle x^2 \rangle - \langle x \rangle^2$$

$\langle x \rangle = dt$ $\qquad$ $\langle x^2 \rangle = (dt)^2 + dt$ $\qquad$ p. 43 of Kardar

so $\langle x^2 \rangle - \langle x \rangle^2 = (dt)^2 + dt - (dt)^2 = dt$

(10) $I = I_0\left[e^{\frac{eV}{kT}} - 1\right]$

$$P(V) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-V^2/2\sigma^2}$$

Using notation from the book $F = I$, $x = V$

equation (2.3) becomes

$$P_I(V) = \sum_i P(V_i) \left| \frac{dV}{dI} \right|_{V=V_i}$$ where V_i is solutions of $I = f$

Find $\frac{dV}{dI}$:

$$I = I_0\left[e^{\frac{eV}{kT}} - 1\right] \qquad \frac{eV}{kT} = \ln\left(\frac{I}{I_0} + 1\right) \qquad V = \frac{kT}{e}\ln\left(\frac{I}{I_0} + 1\right)$$

$$\frac{dV}{dI} = \frac{kT}{e} \frac{1}{\frac{I}{I_0} + 1} \frac{1}{I_0} = \frac{kT}{e(I + I_0)}$$

Note: V is not in $\frac{dV}{dI}$ so no need to evaluate at $V = V_i$

Then $$P_I(V) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-V^2/2\sigma^2} \frac{kT}{e(I + I_0)}$$

Substitute in for V to get

$$P(I) = \frac{kT}{e(I + I_0)\sqrt{2\pi\sigma^2}} \exp\left\{ \frac{-\left[\frac{kT}{e}\ln\left(\frac{I}{I_0} + 1\right)\right]^2}{2\sigma^2} \right\}$$

(10) cont...

most probable value for I?

maximize $P(I)$: $\frac{dP}{dI} = 0$

let $\frac{kT}{e\sqrt{2\pi\sigma^2}} = a$ and $\frac{k^2T^2}{2e^2\sigma^2} = b$

Then $P(I) = \frac{a}{(I+I_0)} \exp\left\{-b \ln^2\left(\frac{I}{I_0}+1\right)\right\}$

$$\frac{dP}{dI} = \frac{a}{(I+I_0)} \exp\left\{-b\ln^2\left(\frac{I}{I_0}+1\right)\right\}\left(-2b\ln\left(\frac{I}{I_0}+1\right)\right)\frac{1}{(I+I_0)}$$

$$+ \frac{(-a)}{(I+I_0)^2} \exp\left\{-b\ln^2\left(\frac{I}{I_0}+1\right)\right\} = 0$$

exponentials cancel, as do $\frac{1}{(I+I_0)^2}$

$$-2ab\ln\left(\frac{I}{I_0}+1\right) = a \qquad \frac{I}{I_0}+1 = e^{-1/2b}$$

$$I = I_0\left(e^{-1/2b}-1\right) = I_0\left(\exp\left\{-\left(\frac{e\sigma}{kT}\right)^2\right\}-1\right)$$

mean value $\langle I \rangle$

$$\langle I \rangle = a\int_{-I_0}^{\infty} \frac{I}{(I_0+I)} \exp\left\{-b\ln^2\left(\frac{I}{I_0}+1\right)\right\} dI \qquad \text{let } I = xI_0,\ dI = I_0\,dx$$

$$\langle I \rangle = aI_0\int_{-1}^{\infty} \frac{x}{x+1} \exp\left\{-b\ln^2(x+1)\right\} dx \qquad \text{now let } 1+x = e^y,\ dx = e^y dy$$

$$\langle I \rangle = aI_0\int_{-\infty}^{\infty} \frac{e^y-1}{e^y} e^y dy\, e^{-by^2} = aI_0\int_{-\infty}^{\infty} dy\,(e^y-1)\,e^{-by^2}$$

$$\langle I \rangle = aI_0\int_{-\infty}^{\infty} e^{-by^2+y}\,dy - aI_0\int_{-\infty}^{\infty} e^{-by^2}dy \qquad \text{Gaussian integrals}$$

$$\langle I \rangle = aI_0\, e^{1/4b}\sqrt{\frac{\pi}{b}} - I_0 a\sqrt{\frac{\pi}{b}} = a\sqrt{\frac{\pi}{b}}\left(I_0 e^{1/4b} - I_0\right) = I_0\left(\exp\left\{\left(\frac{\sigma e}{\sqrt{2}\,kT}\right)^2\right\}-1\right)$$

WolframAlpha computational knowledge engine

```
FindMaximum[(1/(1+x))*Exp[-Log[1+x]^2],{x,-0.5}]
```

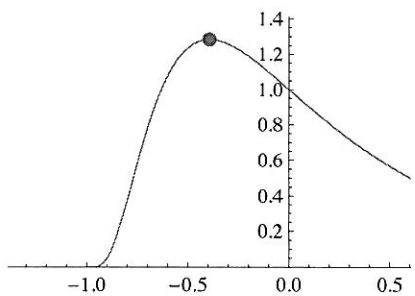
Input interpretation:

local maximum	function	$\frac{e^{-\log^2(1+x)}}{1+x}$
	near	-0.5

Local maximum near x=−0.5:

x	$\frac{e^{-\log^2(x+1)}}{x+1}$
-0.393469	1.28403

Plot:



(x from −1 to 1)

Generated by WolframAlpha (www.wolframalpha.com) on January 28, 2010 from Champaign, IL
© Wolfram Alpha LLC – A Wolfram Research Company